
Multimedia

Presentation of lectures

Department of Computer Graphics and Multimedia



Lecture topic

DSP structures, FIR, IIR

How does the digital part of DSP system work?

Contents:

The basic DSP structures

FIR filter structure

IIR filter structure

More general structures

FIR filter design

IIR filter design

The basic DSP structures Z-transform

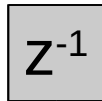
The basic mean of representation of features of DSP elements is the transfer function; as the DSP systems exploit discrete (in time) signal representation, the most suitable is to express the transfer functions in Z-form, eg.:

$$H(z) = b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots + b_N z^{-N}$$

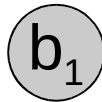
The basic DSP structures elements

The building elements for DSP structures are:

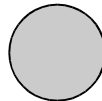
unit delay



multiplication
with constant

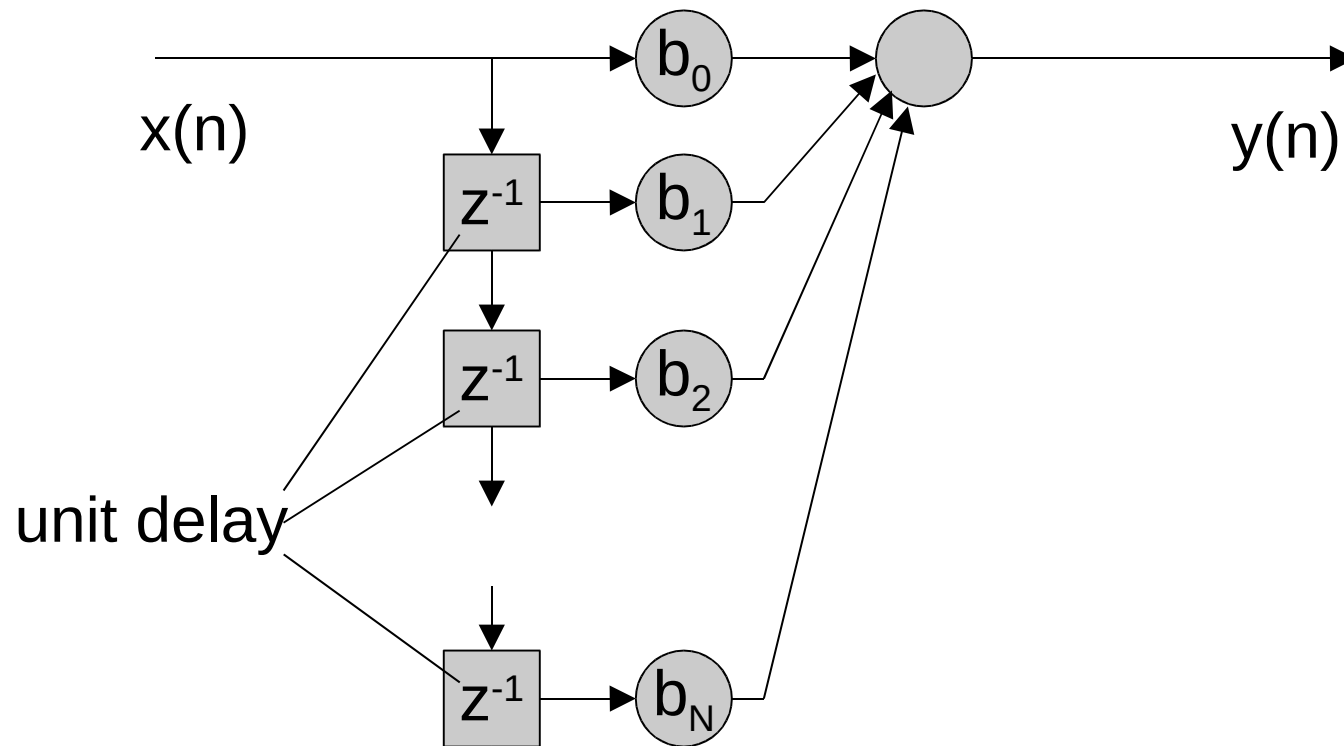


Summing



Filter structures FIR

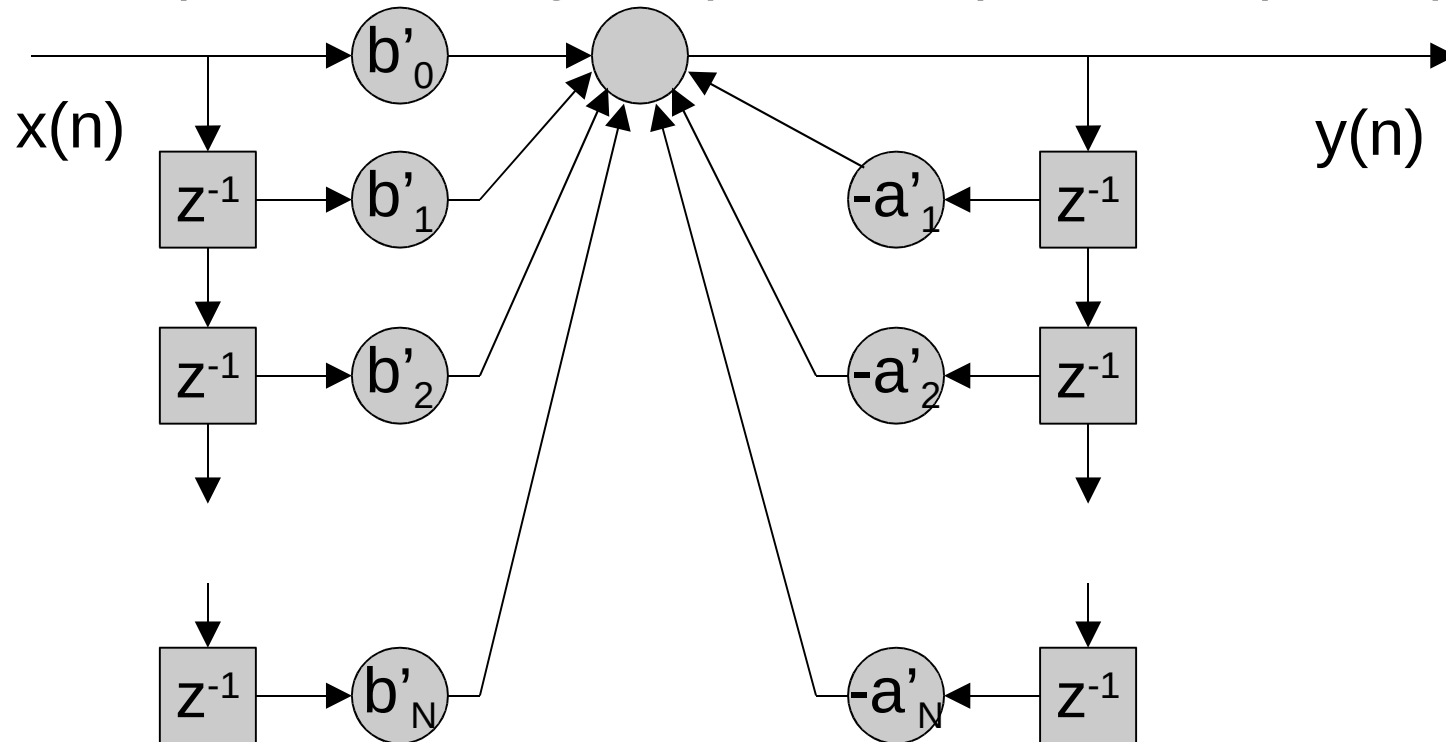
The FIR (Finite Impulse Response) systems are systems whose response to unit impulse is limited in time



$$y[n] = x[n]b_0 + x[n-1]b_1 + \dots + x[n-N]b_N$$

Filter structures IIR

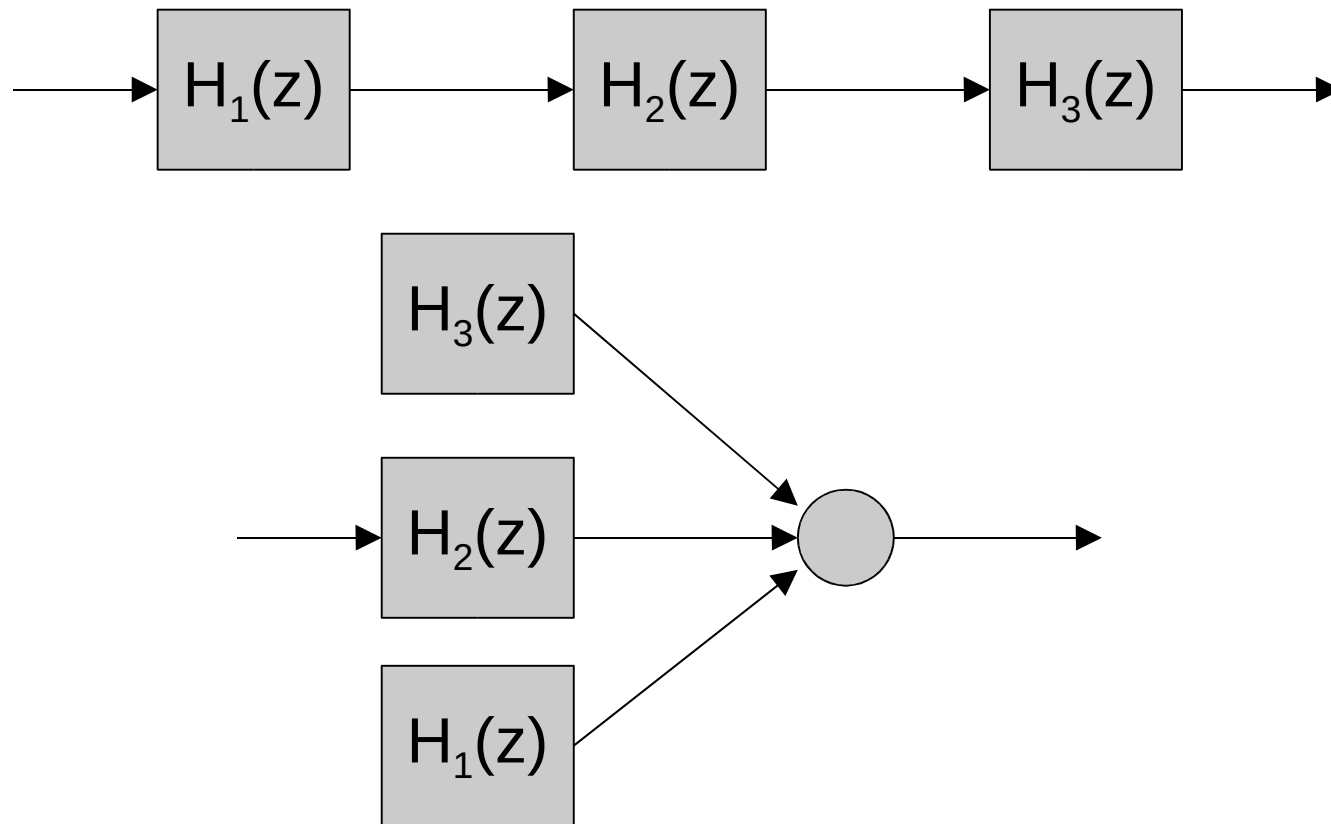
Some more general systems (even with unlimited response) can be expressed using IIR (Infinite Impulse Response)



$$Y(z) = X(z) b'_0 + X(z) b'_1 z^{-1} + \dots + X(z) b'_N z^{-N} - Y(z) a'_1 z^{-1} - Y(z) a'_2 z^{-2} - \dots - Y(z) a'_M z^{-M}$$

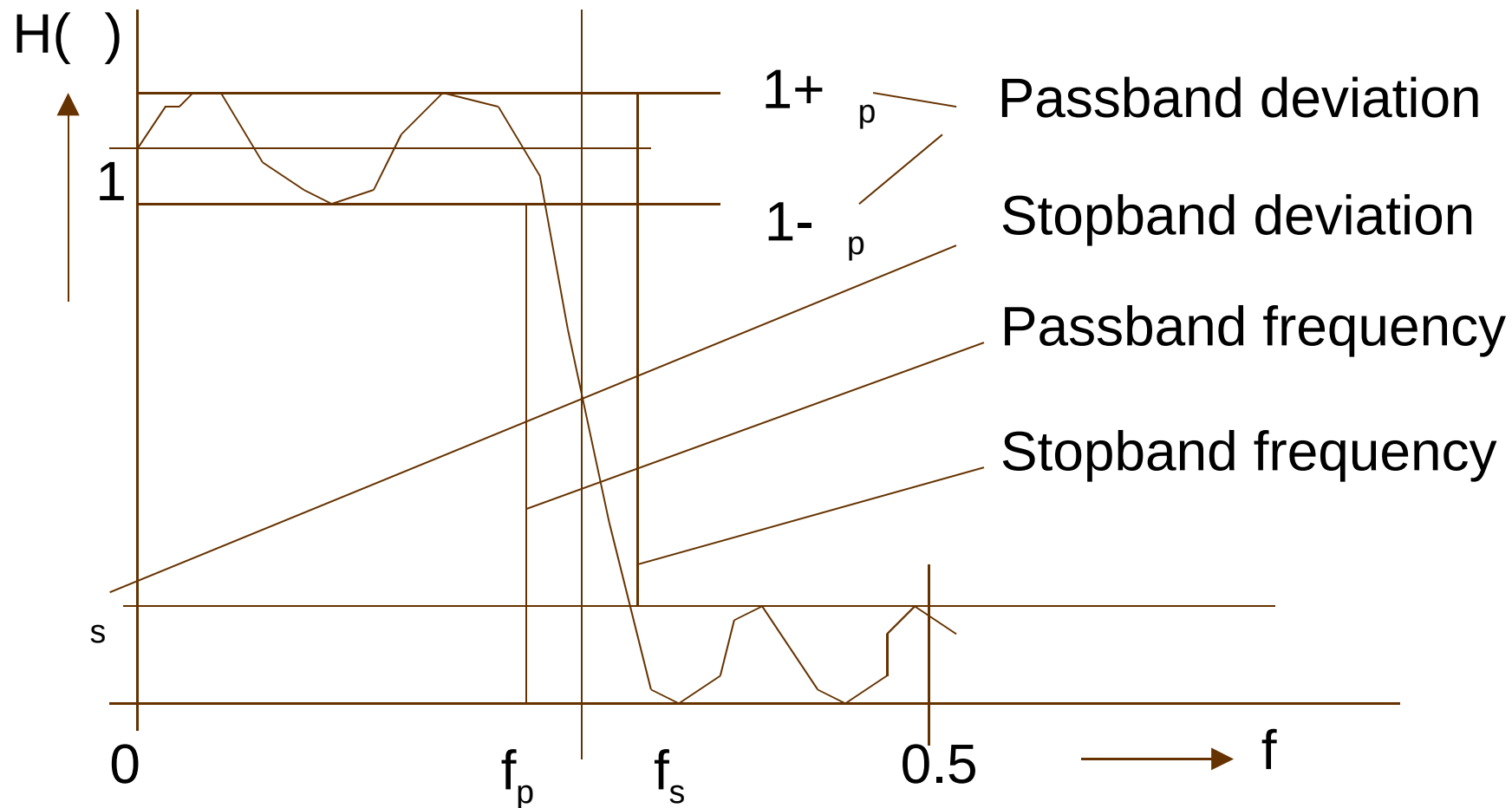
More DSP structures

The DSP structures can be generally built as parallel or serial combination of DSP elements



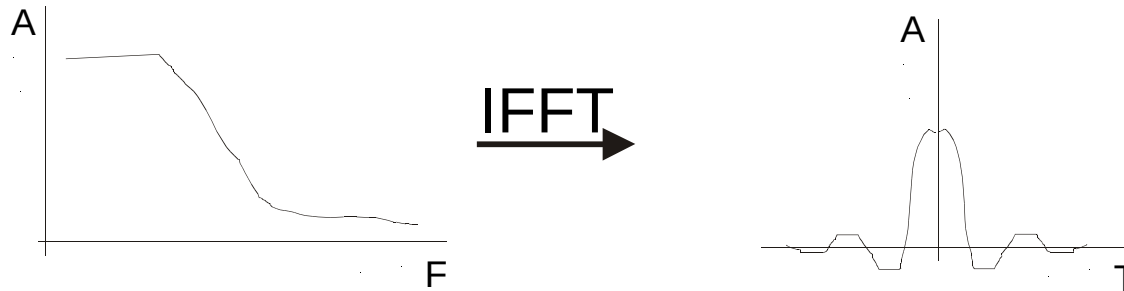
FIR filter design

Example of typical filter characteristics - low pass



FIR filter design

FIR filter design is based on knowledge of the impulse response of the system; if (typically) only frequency response is known, it is necessary to convert it this is usually done through Inverse Fourier Transform



Note: see the FIRComplete.sxc spreadsheet for a complete FIR design from frequency characteristics

FIR filter design

The number of the nonzero coefficients around 0 in time is called the FIR window ; the maximum size of such window is limited by the computational power of the DSP part; the size of the window can be reduced but the problems are:

The impulse response of ideal filters is infinite in time

We can just truncate the impulse response but it introduces error (spectrum of filter combined with spectrum of rectangle defined by the size of nonzero part of window)

Instead of just multiplying the ideal response by rectangle (previous case), we can use (other) window functions to reduce error

FIR filter design

The most frequent window functions are:

Rectangular window $w_n = \begin{cases} 1 & -N/2 \leq n \leq N/2 \\ 0 & \text{elsewhere} \end{cases}$

Hamming window $w_n = \begin{cases} 0.54 - 0.46 \cos \frac{2\pi n}{N} & -N/2 \leq n \leq N/2 \\ 0 & \text{elsewhere} \end{cases}$

Kaiser window $w_n = \begin{cases} I_0 \left(\frac{2\pi n}{N} \sqrt{1 - \beta} \right) / I_0 \left(\frac{2\pi n}{N} \sqrt{1 - \beta} \right) & -N/2 \leq n \leq N/2 \\ 0 & \text{elsewhere} \end{cases}$

$$w_n = \begin{cases} I_0 \left(\frac{2\pi n}{N} \sqrt{1 - \beta} \right) / I_0 \left(\frac{2\pi n}{N} \sqrt{1 - \beta} \right) & -N/2 \leq n \leq N/2 \\ 0 & \text{elsewhere} \end{cases}$$

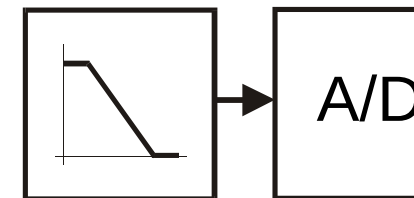
$$I_0(x) = 1 + \sum_{k=1}^L \frac{(x/2)^k}{k!}$$

Note that β allows to finetune stop/pass band

Multirate solution of filtering, A/D, and D/A

One of the main problems in digital signal processing systems is the price of the input filters; fortunately, the complex sharp input filters can be replaced by simple filter that removes frequencies above $N \cdot f_s / 2$, where N is a small number (eg. $N=16$) and preserve frequencies below $f_s / 2$; the output is then digitized and filtered digitally

The consequence is that the input filter can be very simple (eg. RC), the signal must be digitized at higher sample rates (usually no problem for sound) and that the filter must be further digitally filtered (just a small additional computation)



FIR filter design

In most cases, the FIR coefficients are represented in integer type number; this fact causes several problems:

Frequency characteristics change - this is major problem and cannot be solved

DC shift of the filter output - can be overcome by manipulating the coefficients so that their sum is 0 (introduces further freq. error)

Quantization noise - cannot be solved

IIR filter design

No straightforward IIR design method exists; in most cases, design methods known from analog filters are used and then some suitable conversion in IIR is performed, such as:

Pole-zero placement method
(place poles-zeros manually, then get $h(z)$)

Impulse invariant method
(get $H(s)$, convert to $h(z)$ by ILT and Z)

Bilinear Z transform (get $H(s)$
substitute $s = \frac{2}{T} \frac{z-1}{z+1}$, assuming it is linear)

Note that The $H(s)$ is usually obtained by taking suitable standard analog filter design

Conclusions

DSP structures mostly consist of multiplication with constant, multiplication, addition (multiply and accumulate MAC is quite typical), and shift these functions are relatively easy to implement in today's processors; however, very little of functionality of general processors available today is used and the processors can be simple - special DSP processors are being manufactured

FIR filter design is straightforward and simple; IIR structures are more difficult to work with

FIR idea can be extended into 2D; this leads in standard convolution image filters